FinShield – Digital Payment Fraud Behavior Analytics

1. 📌 Introduction

FinShield is a digital payment fraud analytics project designed to analyze transaction data and identify fraudulent patterns. With the rise of online payments, fraud detection has become critical for financial security. This project uses data analytics and visualization techniques to provide insights into fraud behaviour.

Business Objectives:

- Identify high-risk transactions
- Detect unusual spending patterns
- Analyze User behavior shifts
- Build fraud risk scoring
- Help businesses reduce financial loss

2. 📊 Dataset Description

Dataset Includes:

- Transaction ID
- User ID
- Transaction Amount
- Payment Method (Card, UPI, etc.)
- Location
- Risk Score
- Fraud Label (0/1)
- Transaction Time

3. 🛠️ Tools & Technologies

- Power BI
- DAX (for calculations)
- Python (Jupyter Notebook)
- Pandas and NumPy
- MySQL

4. 📐 Methodology

1. Data collection and preprocessing
2. Data cleaning and transformation
3. Creating measures (Fraud Rate using DAX)

4. Building interactive visuals
5. Analyzing patterns and generating insights

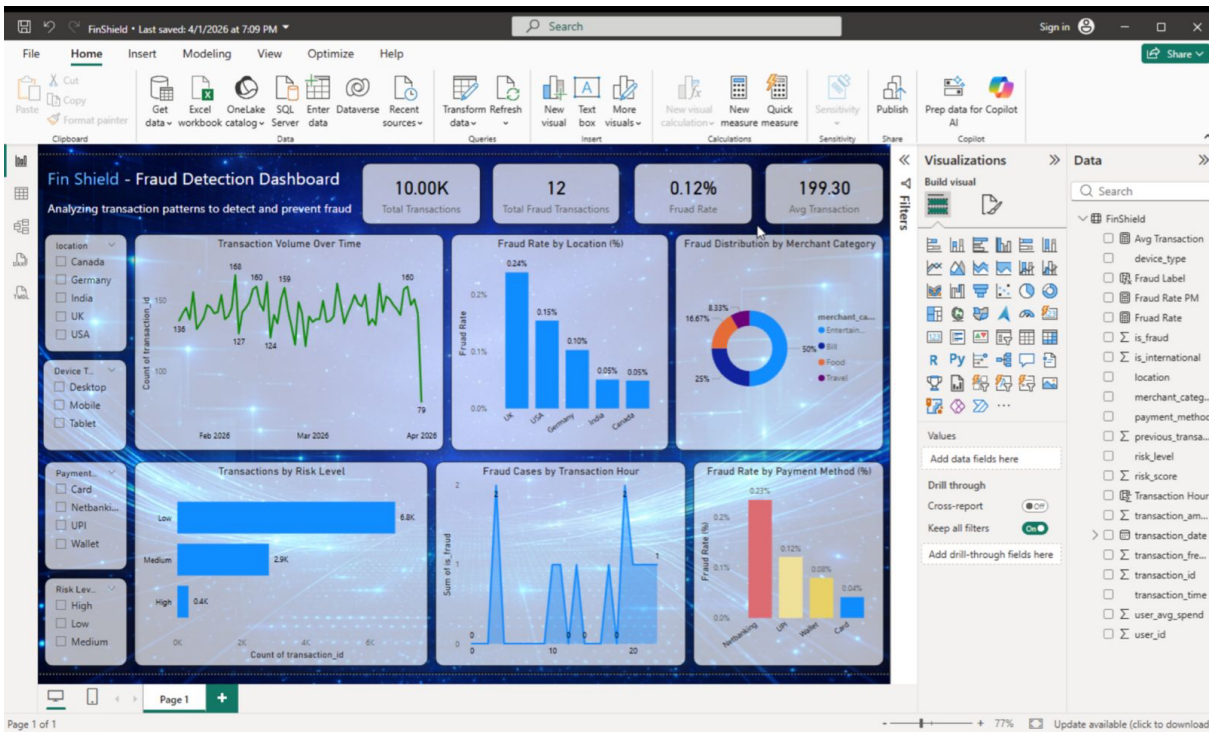
5. 📊Dashboard Overview

The FinShield dashboard is an interactive Power BI dashboard designed to visualize transaction data and identify fraud patterns effectively.

It provides a comprehensive view of key metrics such as total transactions, fraud cases, fraud rate, and average transaction value.

The dashboard allows users to analyze fraud behaviour across different dimensions like time, location, payment methods and risk levels.

6. 📈Dashboard Features



The dashboard consists of multiple interactive visuals that help in analyzing fraud patterns:

- **KPI Cards:**
  - Total Transaction
  - Total Fraud Transactions
  - Overall Fraud Rate (%)
  - Average Transaction Value

- **Time Analysis:**
  - Line Chart showing **Transaction Volume over Time**
- **Location Analysis:**
  - Bar Chart displaying **Fraud Rate by Location (%)**
- **Merchant Analysis:**
  - Donut Chart showing **Fraud Distribution by Merchant Category**
- **Risk Analysis:**
  - Bar Chart representing **Transactions by Risk Level**
- **Time-based Fraud Analysis:**
  - Line Chart showing **Fraud Cases by Transaction Hour**
- **Payment Method Analysis:**
  - Bar Chart showing **Fraud Rate by Payment Method (%)**
- **Filter (Slicers)**
  - Location
  - Device Type
  - Payment Method
  - Risk Level

7. 🔍**Key Insights**
   - Fraud rate is higher in certain locations such as UK
   - Net banking & UPI shows higher fraud rates
   - Fraud activity peaks during specific hours
   - Most transactions are low-risk, but high-risk transactions show higher fraud concentration
   - Certain merchant categories contribute more to fraud
8. 💼Business Impact

   → This dashboard helps businesses identify high-risk transactions and patterns, enabling proactive fraud detection.

   → It reduces financial losses and improve transaction security.